

Report: Optimising NYC Taxi Operations

EDA_Optimising_NYC_Taxis_Arnab Kanu

1. Data Preparation

1.1. Loading the dataset

The 2023 NYC Yellow Taxi dataset consists of 12 monthly Parquet files. To manage computational load, a 5% sample fraction was applied to each hour of every date across all monthly files. This approach preserves the temporal structure of the data while reducing the dataset to a manageable size. All monthly samples were combined into a single analytical dataset and saved as a Parquet file for reuse.

Please Maintain a folder structure according to this path inside the downloads folder of the pc:

C:\Users\username\Downloads\NYC_Taxi_EDA\data: This folder should contain all the parquet files

C:\Users\username\Downloads\NYC_Taxi_EDA\taxi_zones: This should contain all the shape files

C:\Users\username\Downloads\NYC_Taxi_EDA: This folder should contain the IPYNB files

Note: The “username” should be replaced with the systems username

1.1.1. Sample the data and combine the files

By sampling 5% of trips from each date-hour interval, temporal sampling bias was minimised compared to simple random sampling. This ensured the resulting dataset maintained representative traffic demand, revenue distributions and spatial trip patterns, supporting robust operational analysis.

- Sample fraction: 5% per hour per date
- Files processed: 12 monthly Parquet files (January to December 2023)
- Combined dataset saved as: nyc_taxi_2023_sampled.parquet
- random_state=42 used to ensure reproducibility of sampling

2 Data Cleaning

2.1. Fixing Columns

Schema consistency issues were corrected, duplicate columns were consolidated and monetary fields were standardised.

2.1.1. Fix the index

The index introduced during file loading and concatenation was reset to ensure a clean sequential dataframe structure starting from 0.

2.1.2. Combine the two airport_fee columns

The dataset contained two airport fee columns — `airport_fee` and `Airport_fee` — due to a naming inconsistency across monthly files. These were merged into a single `airport_fee` column by filling missing values in `airport_fee` with values from `Airport_fee`, then dropping the duplicate column to avoid double counting.

2.1.3. Fix columns with negative monetary values

Several monetary columns contained negative values which are not realistic for taxi fares. All negative values in `fare_amount`, `extra`, `mta_tax`, `tip_amount`, `tolls_amount`, `improvement_surcharge`, `total_amount`, `congestion_surcharge` and `airport_fee` were converted to their absolute positive equivalents.

2.2. Handling Missing Values

Missing operational fields were handled using context-aware imputations.

2.2.1. Find the proportion of missing values in each column

Highest missingness appeared in `passenger_count`, `RatecodeID`, `congestion_surcharge` and the duplicate airport fee fields.

2.2.2. Handling missing values in `passenger_count`

Missing `passenger_count` values were filled with the median value. The median was chosen over the mean as it is not affected by outliers. Additionally, zero passenger counts were replaced with the median since a taxi trip must have at least one passenger.

2.2.3. Handle missing values in RatecodeID

Missing RatecodeID values were filled with 1.0 (Standard Rate) as this is the most common rate type in the dataset and the safest assumption for unrecorded trips.

2.2.4. Impute NaN in congestion_surcharge

Missing congestion_surcharge values were imputed with 0, indicating no surcharge was applied. This is consistent with the data dictionary which specifies the surcharge is only triggered for specific metered rate types.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

Outlier Rule	Action
passenger_count > 6	Removed — NYC yellow taxis have a maximum capacity of 6
trip_distance > 250 miles	Removed — unrealistic for any NYC taxi trip
fare_amount > 500	Removed — extreme outlier beyond realistic NYC fare range
trip_distance=0 AND fare=0 AND zones different	Removed — invalid trip record
payment_type = 0	Removed — not a valid payment type per data dictionary
trip_duration <= 0	Removed — a trip must have a positive duration

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

Categorical: VendorID, payment_type, RatecodeID, PULocationID, DOLocationID, pickup_hour, pickup_day, pickup_month.

Numerical: fare_amount, tip_amount, trip_distance, passenger_count, all surcharges, total_amount, trip_duration.

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

Demand rises steadily from early morning, peaks between 5PM and 9PM and remains elevated into the evening. Weekdays show a stronger commute-driven structure with morning and evening peaks. Weekends shift toward later-day demand concentrated in the afternoon and evening hours.

3.1.3. Filter out the zero/negative values in fares, distance and tips

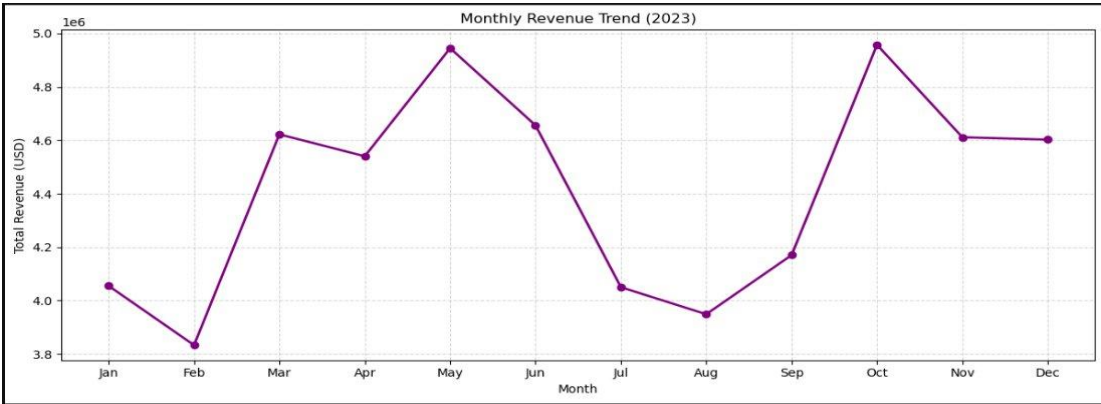
Zero and invalid fare_amount and total_amount values were excluded for financial analysis. tip_amount and trip_distance can legitimately be zero and were retained in the main dataset.

3.1.4. Analyse the monthly revenue trends

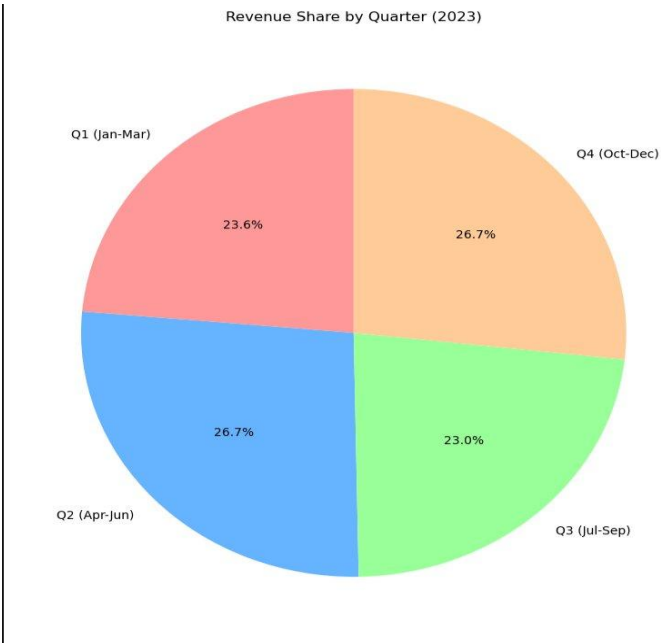
Month	Approximate Revenue (USD)
January	4055112.28
February	3833376.56
March	4622318.81
April	4540408.47
May	4944849.25
June	4656140.21
July	4050211.27
August	3949080.80
September	4170482.88
October	4957562.43

Month	Approximate Revenue (USD)
November	4611506.65
December	4602649.15

Monthly sampled revenue ranged from approximately \$3.85M (February) to \$4.95M (October), showing clear seasonality with summer dips and autumn peaks.

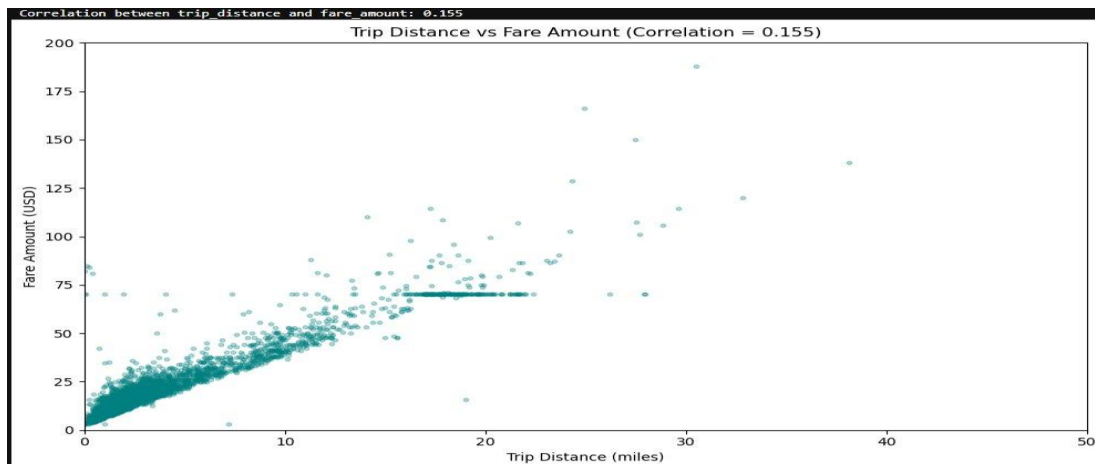


3.1.5. Find the proportion of each quarter's revenue in the yearly revenue
Q1: 23.6%, Q2: 26.7%, Q3: 23.0%, Q4: 26.7%. Revenue concentration is strongest in Q2 and Q4, with Q3 showing the lowest share due to the summer dip in July and August.



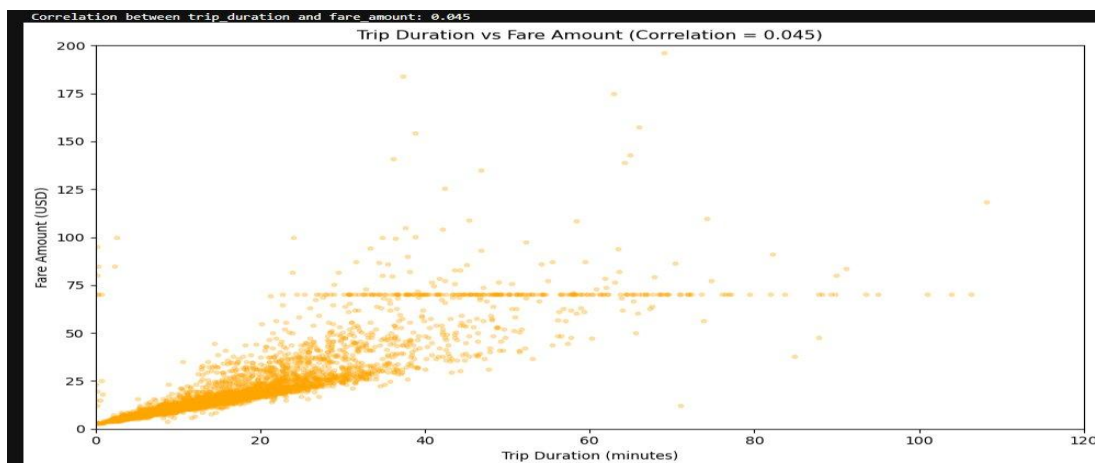
3.1.6. Analyse and visualise the relationship between distance and fare amount

A positive correlation exists between trip distance and fare amount (Correlation = 0.155). While the general trend confirms that longer trips cost more, the moderate correlation value reflects the influence of traffic conditions, rate codes and fixed charges which can cause fares to vary even for similar distances.

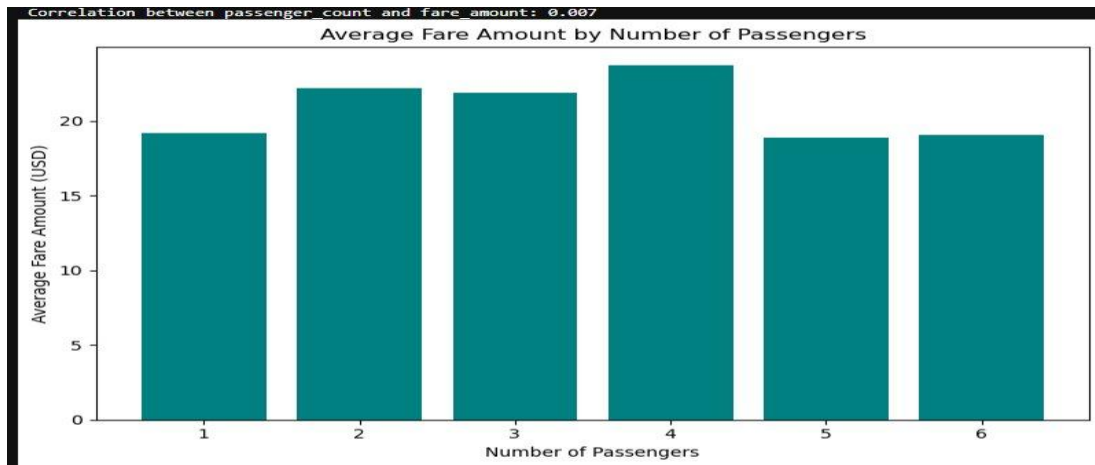


3.1.7. Analyse the relationship between fare/tips and trips/passengers

Fare vs Trip Duration (Correlation = 0.045): The very low correlation suggests that trip duration alone is not a strong predictor of fare amount. This may be because fares in NYC are calculated based on both time and distance.



Fare vs Passenger Count (Correlation = 0.007): Essentially no correlation exists between passenger count and fare amount. This confirms that NYC yellow taxis charge the same fare regardless of the number of passengers.



3.1.8. Analyse the distribution of different payment types

Credit Card payments dominate transactions, suggesting a strong digital payment preference and providing stronger tip capture opportunities. Cash remains the second most common method.

Payment Type	Description	Count
1	Credit Card	1492077
2	Cash	315644
3	No Charge	8878
4	Dispute	13569

3.1.9. Load the taxi zones shapefile and display it

The NYC taxi zone shapefile was loaded using GeoPandas for spatial analysis. The shapefile contains geometric boundaries for all 263 taxi zones across the five NYC boroughs.

3.1.10. Merge the zone data with trips data

Zone data was merged with trip records using LocationID and PULocationID columns, adding zone names and borough information to each trip record for geographical analysis.

3.1.11. Find the number of trips for each zone/location ID

Trip counts were aggregated by pickup zone to identify demand hotspots across the city.

3.1.12. Add the number of trips for each zone to the zones dataframe

Spatial demand metrics were appended to the zones GeoDataFrame for choropleth visualisation.

3.1.13. Plot a map of the zones showing number of trips

Geospatial mapping highlighted the Manhattan core and airport zones as the dominant demand centres. JFK Airport, Upper East Side and Midtown are the top pickup zones.

Rank	Zone	Borough	Approx Trips
1	JFK Airport	Queens	96890.0
2	Upper East Side South	Manhattan	86878.0
3	Midtown Center	Manhattan	85914.0
4	Upper East Side North	Manhattan	77494.0
5	Midtown East	Manhattan	65619.0
6	LaGuardia Airport	Queens	64244.0
7	Penn Station/Madison Sq West	Manhattan	63444.0
8	Times Sq/Theatre District	Manhattan	61289.0
9	Lincoln Square East	Manhattan	60861.0

3.1.14. Conclude with results

Peak demand clusters around commute periods, commercial hubs, entertainment districts and airports. Revenue is seasonally variable but stable overall, with Q2 and Q4 contributing the most to yearly revenue.

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

Routes through central Manhattan exhibited the lowest average speeds, particularly during rush hours (8AM-9AM and 5PM-7PM). These congestion-heavy routes reduce vehicle productivity and increase trip duration without proportional fare increases.

3.2.2. Calculate the hourly number of trips and identify the busy hours

The busiest hours are between 5PM and 9PM. Demand begins rising steadily from midday, peaks in the early evening and gradually declines after 9PM.

3.2.3. Scale up the number of trips from above to find the actual number of trips

Since a 5% sample was taken, the actual number of trips is estimated by dividing sampled counts by 0.05 (multiplying by 20). The busiest hours are estimated to handle millions of trips across NYC.

3.2.4. Compare hourly traffic on weekdays and weekends

Weekdays show a clear double-peak pattern driven by morning commute (7-9AM) and evening commute (5-9PM). Weekends show a single broad peak in the afternoon and evening (12PM-11PM) with no morning rush, driven by leisure and social activities.

3.2.5. Identify the top 10 zones with high hourly pickups and drops

JFK Airport, LaGuardia, Midtown East, Times Square, Penn Station and Upper East Side zones consistently rank in the top 10 for both pickups and dropoffs.

3.2.6. Find the ratio of pickups and dropoffs in each zone

Zones with high pickup-to-dropoff ratios are natural origin points for taxis. Zones with low ratios are destination-heavy and represent opportunities to reposition taxis after dropoffs.

Type	Location ID/Zone	Ratio
Highest (more pickups)	70/ East Elmhurst	8.43
Highest (more pickups)	128/ JFK Airport	4.63
Highest (more pickups)	134/ LaGuardia Airport	2.89

Type	Location ID/Zone	Ratio
Lowest (more dropoffs)	99/ Freshkills Park	0.00
Lowest (more dropoffs)	29/ Broad Channel	0.00
Lowest (more dropoffs)	172/ Oakwood	0.00

3.2.7. Identify the top zones with high traffic during night hours

Airport, nightlife and entertainment zones dominate 11PM to 5AM demand. Manhattan entertainment districts such as Hell's Kitchen, Midtown and Times Square show the highest night traffic.

3.2.8. Find the revenue share for nighttime and daytime hours

Daytime hours (6AM-10PM) contribute the larger share of total revenue. However nighttime hours (11PM-5AM) remain strategically important due to concentrated demand in entertainment zones and higher fare per mile during overnight hours.

3.2.9. For the different passenger counts, find the average fare per mile per passenger

Single-passenger trips generate the highest fare per mile per passenger. As passenger count increases, the per-passenger fare per mile decreases since the fare remains fixed while being shared among more passengers.

3.2.10. Find the average fare per mile by hours of the day and by days of the week

Fare per mile is highest during early morning hours (2AM-5AM) partly due to the overnight surcharge. Weekend days show slightly higher fare per mile compared to weekdays, possibly due to longer trips and more airport runs.

3.2.11. Analyse the average fare per mile for the different vendors

Fare efficiency varies by time and vendor, reflecting differences in congestion exposure and trip mix. Both Vendor 1 (Creative Mobile Technologies) and Vendor 2 (VeriFone) show similar overall patterns but with modest differences at specific hours.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

Vendor performance differs modestly across distance tiers, suggesting benchmarking opportunities. For short trips (under 2 miles), differences are more pronounced. For longer trips (5+ miles), rates converge.

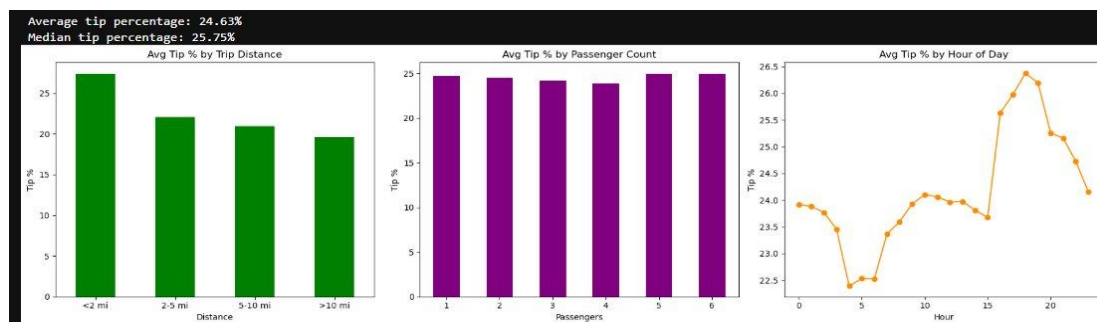
3.2.13. Analyse the tip percentages

Distance Tier	Average Tip %
< 2 miles	28%
2-5 miles	23%
5-10 miles	20%
> 10 miles	19%

Average tip percentage: 24.63%

Median tip percentage: 25.75%

Short trips (under 2 miles) show the strongest tip percentages, declining with trip length. Tips are fairly consistent across passenger counts and peak during mid-afternoon to evening hours.



3.2.14. Analyse the trends in passenger count

Single riders dominate overwhelmingly. Multi-passenger trips are significantly fewer. Average passenger count stays around 1.3-1.6 per trip throughout the day and week. Weekend evenings show slightly higher average passenger counts due to groups travelling together.

3.2.15. Analyse the variation of passenger counts across zones

Commercial and airport zones show different occupancy behaviour compared to residential areas. Airport zones tend to have slightly higher average passenger counts as travellers often share rides. Central Manhattan zones predominantly serve solo commuters.

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently

Surcharge frequency observed in sampled data:

Surcharge Type	Trips Affected	Notes
extra (rush/overnight)	1,100,869	Applied during rush hours and overnight
mta_tax	1,454,975	Applied to almost all standard metered trips
improvement_surcharge	1,554,149	Applied at flag drop for almost all trips
congestion_surcharge	1,486,506	Applied in central Manhattan zones
airport_fee	152,163	Applied only at JFK and LaGuardia pickups

4 Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies

- Deploy more taxis during peak hours (5PM-9PM) on weekdays to meet the highest demand periods
- Use traffic-aware routing to avoid slow central Manhattan routes during rush hours (8-9AM and 5-7PM)
- Implement dynamic dispatch that responds to real-time demand signals from high-pickup zones
- Correct pickup-dropoff imbalances by redirecting taxis from low-pickup sink zones to nearby high-demand zones after dropoffs
- On weekends, shift deployment from morning to afternoon as demand peaks later in the day
- Maintain night coverage in entertainment zones (Hell's Kitchen, Midtown, Times Square) from 10PM onwards on Fridays and Saturdays

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months

- Pre-position taxis in Manhattan business districts (Midtown, Upper East Side) during commute windows (7-9AM and 5-7PM)
- Maintain continuous airport coverage at JFK and LaGuardia as these generate the highest trip volumes and longer revenue trips
- Position taxis in nightlife zones overnight on weekends to capture concentrated late-night demand
- Rebalance vehicles from destination-heavy zones (low pickup ratio) to origin-heavy zones (high pickup ratio) to reduce idle time
- Use the choropleth map to identify underserved zones and strategically expand coverage

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors

- Apply moderate surge pricing during early morning (2-5AM) and peak evening (5-9PM) hours when demand is highest
- Introduce a minimum fare for very short trips (under 0.5 miles) to ensure operational costs are covered

- Encourage credit card payments as they capture tip data and contribute to higher overall revenue — 24.63% average tip rate
- Benchmark pricing against both vendors across distance tiers to identify competitive pricing opportunities
- For airport routes, maintain the current flat rate structure as passengers expect and accept these rates
- Consider incentivising off-peak trips with slightly lower fares to spread demand more evenly and improve fleet utilisation

• Assumptions Made

- A 5% random sample per hour per date is representative of overall trip distribution
- Negative monetary values are data entry errors — converting to positive is appropriate
- Missing passenger_count values can be filled with the median value
- Missing RatecodeID values can be filled with 1.0 (Standard Rate)
- Missing congestion_surcharge and airport_fee values indicate no charge was applied (filled with 0)
- Trip distance of 0 with the same pickup and dropoff zone is a valid trip
- Fare amounts above \$500 are data errors and treated as outliers
- Only credit card tips are available — cash tips are not recorded
- The two airport_fee columns represent the same field with a naming inconsistency

• Tools and Libraries Used

- Python 3 — Programming language
- Pandas 2.3.3 — Data manipulation and analysis
- NumPy 2.3.5 — Numerical calculations
- Matplotlib 3.10.6 — Data visualisation
- Seaborn 0.13.2 — Statistical charts
- GeoPandas — Geographical analysis and choropleth mapping
- Jupyter Notebook (Anaconda) — Development environment